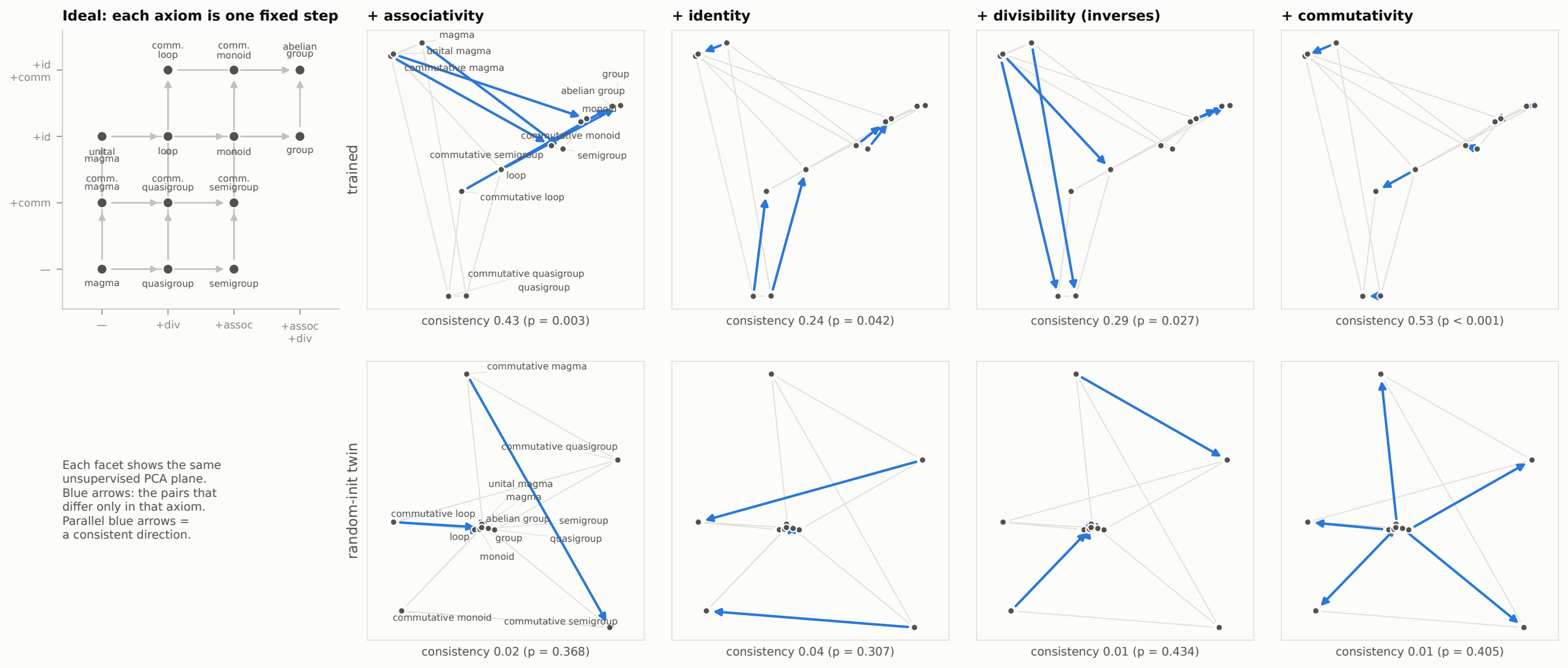
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
trained
random-init twin
+id +comm
+id
+comm
—
comm. loop
comm. monoid
abelian group
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
magma
quasigroup
semigroup
— +div +assoc +assoc +div
magma
unital magma
commutative magma
group
abelian group
monoid
commutative monoid
commutative semigroup
semigroup
loop
commutative loop
commutative quasigroup
quasigroup
consistency 0.43 (p = 0.003)
consistency 0.24 (p = 0.042)
consistency 0.29 (p = 0.027)
consistency 0.53 (p < 0.001)
commutative magma
commutative quasigroup
unital magma
magma
commutative loop
abelian group
semigroup
loop
group
quasigroup
monoid
commutative monoid
commutative semigroup
consistency 0.02 (p = 0.368)
consistency 0.04 (p = 0.307)
consistency 0.01 (p = 0.434)
consistency 0.01 (p = 0.405)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.